



Configure terminal session recording

Lab by Justesse Louboulat Milandou

Why this lab ?

Implementing a terminal session recording lab is essential to validate the traceability and compliance of privileged access before any production deployment. This lab will allow us to test the raw capture of administrator activity (screen input/output), making any history tampering or circumvention via sudo impossible to conceal—which is a key requirement of security standards like PCI-DSS or ISO 27001. Furthermore, this experimental phase is crucial to measure the real technical impact of the solution: it gives us the opportunity to configure log centralization toward, journald anticipate the required storage space, and verify the absence of user-side latency, thereby avoiding any disruption to the company's critical infrastructure.

Prerequisites For this lab :

- Be on Red hat enterprise linux system
 - Install cockpit-session-recording and tlog
-

Roles of packets

Cockpit-session-recording : visual graphic modul for cockpit interface web.

Tlog : This tool capture in background and recording the flow (entrées/sorties) off terminal

```

/usr/bin/cockpit-session-recording
milandou@localhost:~$ sudo dnf install cockpit-session-recording -y
[sudo] password for milandou:
Updating Subscription Management repositories.
Last metadata expiration check: 7:37:37 ago on Tue 11 Aug 2026 02:47:35 PM GMT.
Dependencies resolved.
=====
Package                Arch    Version      Repository      Size
=====
Installing:
cockpit-session-recording
noarch 20-1.el10  rhel-10-for-x86_64-appstream-rpms 580 k
Installing dependencies:
tlog                x86_64 14-4.el10  rhel-10-for-x86_64-appstream-rpms 124 k

Transaction Summary
=====
Install 2 Packages

Total download size: 705 k
Installed size: 860 k
Downloading Packages:
(1/2): tlog-14-4.el10.x86_64.rpm          172 kB/s | 124 kB    00:00
(2/2): cockpit-session-recording-20-1.el10.no 649 kB/s | 580 kB    00:00
=====

```

We can remark when we install **cockpit-session-recording**, it install also his dependance : **tlog**.

milandou
localhost

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Session Recording Settings

General Config

Shell /bin/bash

Notice ATTENTION! Your sess...

Latency 10

Payload Size, bytes 2048

Logging
☐ User's Input
☒ User's Output
☒ Window Resize

Limit Rate, bytes/sec 16384

Burst, bytes 32768

Logging Limit Action Pass

File Path

Syslog Facility authpriv

Syslog Priority Info

Journal Priority Info

SSSD Config

Scope None

All

Open navigator and open cockpit interface at **localhost:9090**, and go to the **session recording** tab

In config session recording i change the scope to all Through gestionnary **SSSD**, for recording all users on our system. The others parameters it stays default.

```
milandou@localhost:~$ cat /etc/tlog/tlog-rec-session.conf
{"shell":"/bin/bash","notice":"\nATTENTION! Your session is being recorded!\n\n","latency":10,"payload":1000,"window":true,"limit":{"rate":16384,"burst":32768,"action":"pass"},"file":{"path":"","syslog":{"journal":{"priority":"info","augment":true},"writer":"journal"}}milandou@localhost:~$
```

We can apply others modifications, like notice or latency, in **tlog-rec-session.conf** file

```
milandou@localhost:~$ cat /etc/ss
ssh/  ssl/  sssd/
milandou@localhost:~$ cat /etc/sss/conf.d/sss-session-recording.conf
cat: /etc/sss/conf.d/sss-session-recording.conf: Permission denied
milandou@localhost:~$ sudo cat /etc/sss/conf.d/sss-session-recording.conf
[sudo] password for milandou:
[sss]
services=nss, pam
domains=nssfiles

[domain/nssfiles]
id_provider=proxy
proxy_lib_name=files
proxy_pam_target=sss-shadowutils

[session_recording]
scope=all
exclude_users=none
exclude_groups=
milandou@localhost:~$ _
```

We can also modify SSSD config by editing **sss-session-recordin.conf** file.



ATTENTION! Your session is being recorded!

```
milandou@localhost:~$ exit_
```

Now when we open our terminal we can see :
our **notice** “ATTENTION! Your session is being recorded!”. to prevent the user.

Let's doing some actions, to see these on
session recording and journald service :

```
milandou@localhost:~$ ls /tmp/
AC_drop.{12d34_56n78_90s12_34m56}.png
AC_item.{12d34_56n78_90s12_34m56}.png
'Air Cooler_drop.{12d34_56n78_90s12_34m56}.png'
'Air Cooler_item.{12d34_56n78_90s12_34m56}.png'
Alarm_drop.{12d34_56n78_90s12_34m56}.png
Alarm_item.{12d34_56n78_90s12_34m56}.png
Appliance_drop.{12d34_56n78_90s12_34m56}.png
Appliance_item.{12d34_56n78_90s12_34m56}.png
'Atm Pressure Monitor_drop.{12d34_56n78_90s12_34m56}.png'
'Atm Pressure Monitor_item.{12d34_56n78_90s12_34m56}.png'
banner-web.jpg
Battery_drop.{12d34_56n78_90s12_34m56}.png
Battery_item.{12d34_56n78_90s12_34m56}.png
Beacon_drop.{12d34_56n78_90s12_34m56}.png
Beacon_item.{12d34_56n78_90s12_34m56}.png
Blower_drop.{12d34_56n78_90s12_34m56}.png
Blower_item.{12d34_56n78_90s12_34m56}.png
'Bluetooth Speaker_drop.{12d34_56n78_90s12_34m56}.png'
'Bluetooth Speaker_item.{12d34_56n78_90s12_34m56}.png'
'Carbon Dioxide Detector_drop.{12d34_56n78_90s12_34m56}.png'
'Carbon Dioxide Detector_item.{12d34_56n78_90s12_34m56}.png'
'Carbon Monoxide Detector_drop.{12d34_56n78_90s12_34m56}.png'
```

```
milandou@localhost:~$ who
milandou tty2      2026-08-11 13:44 (local)
milandou pts/1     2026-08-11 22:48
milandou web console 2026-08-11 22:21 (:1)
milandou@localhost:~$ who
milandou tty2      2026-08-11 13:44 (local)
milandou pts/1     2026-08-11 22:48
milandou web console 2026-08-11 22:21 (:1)
milandou@localhost:~$ df -hP
Filesystem          Size  Used Avail Use% Mounted on
/dev/mapper/rhel-root 70G   67G   3.4G  96% /
devtmpfs             16G    0    16G   0% /dev
tmpfs                16G  139M   16G   1% /dev/shm
efivarfs             192K  132K   56K  71% /sys/firmware/efi/efivars
tmpfs                6.2G   18M   6.2G   1% /run
tmpfs                1.0M    0  1.0M   0% /run/credentials/systemd-journald.service
/dev/nvme0n1p2       2.0G  949M   1.1G  49% /boot
/dev/mapper/rhel-home 389G   22G  367G   6% /home
/dev/nvme0n1p1       599M   9.0M  590M   2% /boot/efi
tmpfs                3.1G  268K   3.1G   1% /run/user/1000
milandou@localhost:~$
```


Duration

06:52

[illegible]

01:47 / 07:02

› **Logs View**

We need to go back on cockpit web interface, go to session recording tab and **select our user**, here "milandou"

After you can remark, this recording of milandou's (user) terminal display like video. And contains all moves of the user.

```

milandou@localhost:~$ journalctl -o verbose | grep -P "\rec\".*?\,"
MESSAGE={"ver":"2.3","host":"localhost","rec":"72f0a44217e94fc68f226b0967522d26-196a4-326df7","user":"milandou","term":"xterm-256color","session":3,"id":1
,"pos":0,"time":1786488514.197,"timing":"=141x31+318>77","in_txt":"","in_bin":[],"out_txt":"\u001b[0m;milandou@localhost:~\u0007\u0001b[?2004h\u0001b[32mmilandou
@localhost\u001b[0m;\u001b[32m~\u001b[0m\u001b[0m","out_bin":[]}]
MESSAGE={"ver":"2.3","host":"localhost","rec":"72f0a44217e94fc68f226b0967522d26-196a4-326df7","user":"milandou","term":"xterm-256color","session":3,"id":2
,"pos":58757,"time":1786488572.954,"timing":"=141x31>1+151>1+398>1+595>1+495>1+150>1+200>2+1172>11+7>1639","in_txt":"","in_bin":[],"out_txt":"ls /tmp/\r\n\u00
1b[?2004\r\n\u0001b[0m;\u001b[01;35mAC_drop.{12d34_56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;35mAC_item.{12d34_56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;
35m'Air Cooler_drop.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Air Cooler_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35mAlarm_dr
op.{12d34_56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;35mAlarm_item.{12d34_56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;35mAppliance_drop.{12d34_56n78_90s1
2_34m56}.png\u001b[0m\r\n\u0001b[01;35mAppliance_item.{12d34_56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;35m'Atm Pressure Monitor_drop.{12d34_56n78_90s12_34m
56}.png'\u001b[0m\r\n\u0001b[01;35m'Atm Pressure Monitor_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35mbanner-web.jpg\u001b[0m\r\n\u0001b[01;35
mBattery_drop.{12d34_56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;35mBattery_item.{12d34_56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;35mBeacon_drop.{12d34_
56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;35mBeacon_item.{12d34_56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;35mBlower_drop.{12d34_56n78_90s12_34m56}.png
\u001b[0m\r\n\u0001b[01;35mBlower_item.{12d34_56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;35m'Bluetooth Speaker_drop.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n
\u0001b[01;35m'Bluetooth Speaker_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Carbon Dioxide Detector_drop.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Carbon Dioxide Detector_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Carbon Monoxide Detector_drop.{12d34_56n78_90s1
2_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Carbon Monoxide Detector_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Ceiling Fan_drop.{12d34_56n78_9
0s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Ceiling Fan_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Ceiling Sprinkler_drop.{12d34_56n78_90s12
_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Ceiling Sprinkler_item.{12d34_56n78_9","out_bin":[]}]
MESSAGE={"ver":"2.3","host":"localhost","rec":"72f0a44217e94fc68f226b0967522d26-196a4-326df7","user":"milandou","term":"xterm-256color","session":3,"id":3
,"pos":61925,"time":1786488576.122,"timing":"=141x31>802+1>893","in_txt":"","in_bin":[],"out_txt":"0s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Dimmable LED_dro
p.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Dimmable LED_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35mDoor_drop.{12d34_56n78_9
0s12_34m56}.png\u001b[0m\r\n\u0001b[01;35mDoor_item.{12d34_56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;35mequipment1-web.jpg\u001b[0m\r\n\u0001b[01;35m'Fire
Monitor_drop.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Fire Monitor_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Fire Sprinkl
er_drop.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Fire Sprinkler_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Flex Sensor_drop
.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Flex Sensor_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Floor Sprinkler_drop.{12d3
4_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Floor Sprinkler_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35mFurnace_drop.{12d34_56n78_90
s12_34m56}.png\u001b[0m\r\n\u0001b[01;35mFurnace_item.{12d34_56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;35m'Garage Door_drop.{12d34_56n78_90s12_34m56}.png'\
\u001b[0m\r\n\u0001b[01;35m'Garage Door_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Generic Environment Sensor_drop.{12d34_56n78_90s12_34m56}.
png'\u001b[0m\r\n\u0001b[01;35m'Generic Environment Sensor_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Generic Sensor_drop.{12d34_56n78_90s12
_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Generic Sensor_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;34mgnome-desktop-thumbnailer-2WNZT3\u0001b[0m\r\n
\u0001b[01;35m'Heating Element_drop.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Heating Element_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n
\u0001b[01;35m'Home Speaker_drop.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b[01;35m'Home Speaker_item.{12d34_56n78_90s12_34m56}.png'\u001b[0m\r\n\u0001b
[01;35mHumidifier_drop.{12d34_56n78_90s12_34m56}.png\u001b[0m\r\n\u0001b[01;","out_bin":[]}]
MESSAGE={"ver":"2.3","host":"localhost","rec":"72f0a44217e94fc68f226b0967522d26-196a4-326df7","user":"milandou","term":"xterm-256color","session":3,"id":4

```

The screenshot illustrates raw log session recording data as stored within the RHEL system journal (**journald**)." Cockpit take this and transform that in video

Terminal Session Recording offers absolute and tamper-proof traceability of administrator activity by capturing all screen inputs and outputs, ensuring compliance with strict security standards (PCI-DSS, ISO 27001) and simplifying post-mortem analysis with minimal CPU overhead. On the flip side, this solution generates a significant volume of logs in journald that are unreadable in their raw state (JSON saturated with ANSI codes), requires a robust audit infrastructure, and introduces a data leakage risk if sensitive data or passwords are displayed in clear text on the screen.

In conclusion, implementing Terminal Session Recording in a lab environment confirms that it is an essential pillar for modern infrastructure governance. Although this solution introduces real technical challenges—most notably rigorous storage volume management and the protection of sensitive data displayed on-screen—its security benefits far outweigh them. By guaranteeing tamper-proof traceability against privilege escalation and natively meeting PCI-DSS or ISO 27001 compliance requirements, this tool does more than just monitor: it enforces access accountability and sustainably secures production. This lab thus provides all the keys to orchestrating a controlled, seamless, and risk-free global deployment.

Sources :

<https://zero.rhdp.net/lab/zt-rhelbu.zt-session-recording-tlog.prod>

https://docs.redhat.com/en/documentation/red_hat_enterprise_linux/8/html/recording_sessions/getting-started-with-session-recording_getting-started-with-session-recording

THANKS

Lab by **Justesse Louboulat Milandou**

Second year Cybersecurity